ONE DISTRICT ONE PRODUCT
UTTAR PRADESH

A GUIDE TO PRODUCT QUALITY STANDARDS

UTENSILS

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INDEX

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ABBREVATIONS

ANSI	American National Standards Institute
APLAC	Asia Pacific Laboratory Accreditation Co-operation
AS	Australian Standards
ASTM	American Society for Testing and Materials
BIS	Bureau of Indian Standards
CE	Conformité Européenne
CoC	Certificate of Conformity
CPSC	Consumer Products Safety Commission
CPSIA	Consumer Product Safety Improvement Act
CPSA	Consumer Product Safety Act
EC	Council Regulation (of the European Parliament and Council)
ECHA	European Chemicals Agency
EN	European Standards
EPA	Environment Protection Act
EU	European Union
FCM	Food Contact Materials
FDA	Food and Drug Administration
FHSA	Federal Hazardous Substances Act
FSSAI	Food Safety and Standards Authority of India
GCC	Gulf Co-operation Council
GI	Geographical Indication
GSO	GCC Standardization Organization
GPSD	General Product Safety Directive
IS	Indian Standard
ISI	Indian Standards Institute
ILAC	International Laboratory Accreditation Co-operation
ISO	International Organization for Standardization
MDF	Medium Density Fibre
MLA	Multilateral Recognition Arrangement
MRA	Mutual Recognition Arrangement
NABCB	National Accreditation Board for Certification Bodies
NABL	National Accreditation Board for Testing and Calibration Laboratories
REACH	Registration, Evaluation, Authorization and Restriction of Chemicals
RoHS	Restriction of Hazardous Substances Directive
SASO	Saudi Standards, Metrology and Quality Organization
QCI	Quality Council of India
QCO	Quality Control Order

BACKGROUND

Uttar Pradesh, the Indian state with a geographical expanse of 2, 40,928 sq.km and a population of 204.2 million people is home to a great diversity in all facets of life. With diverse community traditions and economic pursuits, Uttar Pradesh is also endowed with a beautiful diversity of crafts and industries spread across small towns and districts known for the uniqueness of these products.

To encourage such indigenous products and crafts, the Uttar Pradesh government has initiated the One District One Product (ODOP) Scheme with a vision to create product-specific traditional industrial hubs across 75 districts of Uttar Pradesh that will also promote traditional industries that are synonymous with the respective districts of the state. Although many of the district-specific industries are more of a commonplace but there are several products that are unique to those regions.

Many of these products are GI-tagged; a geographical indication (GI) is a sign used on products that have a specific geographical origin and possess qualities or a reputation that are due to that origin. This becomes a testimony to the original creativity and exclusiveness of some of the finest products produced in the various districts of Uttar Pradesh.

To help in achieving the objectives of the ODOP initiative, the ODOP Cell, Government of Uttar Pradesh, engaged Quality Council of India (QCI) to develop 'Compendiums of relevant Regulations and Standards' for the products under this initiative. The compendiums have been developed by the Quality Council of India with assistance of the sectoral and product experts. Field visits and virtual interactions with ODOP manufacturers were also undertaken to assess the level of their understanding on relevant standards and certifications. The ODOP Cell, Government of Uttar Pradesh and the EY team at the ODOP Cell also provided continuous support in the development of these compendiums.

The key objective of developing the compendiums is to provide a ready reference of various regulations & standards for the existing ODOP manufacturers and producers and also for the aspiring entrepreneurs.

This part of the compendium brings relevant regulations and standards pertaining to the utensils produced in various districts of the state, for example, the region of Pukhrayan in Kanpur district is especially famous for its utensils made from both aluminium and brass whereas Sant Kabir Nagar is renowned for its brassware items of utility including utensils and their quality of craftsmanship.

ALUMINIUM UTENSILS, KANPUR DEHAT

1. INTRODUCTION

More than half of the cookware sold each year globally are made of aluminium. The advantage of aluminium over other materials for kitchen utensils lies in its excellent thermal conductivity (which is approximately an order of magnitude greater than that of steel) while offering the dual advantages of being extremely light and largely non-reactive with foodstuffs at low and high temperatures. Aluminium utensils in addition also have very low toxicity.

Aluminium came to India only in 1938 and the first use of the metal in India was to make cooking utensils in a small factory in the vicinity of Calcutta. Till 1970s, the major use of the primary Aluminium produced in India was to make utensils. Almost 30% of the primary metal was consumed to make cookware. With proper care, Aluminium utensils can last a lifetime. Since aluminium has been readily available, consumers, professional chefs and commercial food processors have welcomed its advantages for food preparation. Today, almost all cookware in India is made by recycling aluminium scrap. Recycling aluminium scrap consumes only 5% of the energy used to make the same amount of primary metal.

Aluminium is easily recyclable and therefore, has a good resale value. Good number of industrials units are located in Kanpur Dehat district that are involved in manufacturing of aluminium utensils.

2. RAW MATERIALS

Pure Aluminium is an extremely soft metal and thus Aluminium is almost always alloyed, which markedly improves its mechanical properties, especially when tempered. The main alloying agents are copper, zinc, magnesium, manganese, and silicon (e.g., duralumin) with the levels of other metals in a few percent by weight. For manufacture of utensils aluminium sheets of varying thicknesses are used. Sheet metal is metal formed by an industrial process into thin, flat pieces. Sheet metal is one of the fundamental forms used in metalworking, and it can be cut and bent into a variety of shapes. Countless everyday objects are fabricated from sheet metal.

3. MANUFACTURING PROCESS

Stages involved in the making of aluminium utensils are:

3.1. Ingots or Beams

The manufactured beams are rolled into thin sheets to produce vessels, cans etc.

3.2. Circles or Discs

Discs are cut from the sheet for producing utensils of various shapes and sizes.

3.3. Tooling and lathing

The disc is fixed on spinning lathe and rotated at high speed. The craftsman then applies the pressure by pressing the metal sheet with the help of roller tool (an elongated thick metal rod). The desired shape is formed when the sheet is pressed against a tool called mandrel. Mandrel is a solid from inserted to the lathe machine to obtain the internal shape of the vessel. Different shapes of vessels can be achieved by inserting variety of mandrel forms. The cylindrical shaped vessel is completed in the first stage by using a particular type of lathe. Using a different type of spinning/turning lathe, the vessel is given its final shape as the upper part of the vessel is shaped out of the hollow like container with large opening which was the outcome of the first stage of processing. In the second stage, iron weights or moulds are placed inside the cylindrical vessel and fixed to the lathe machine. The vessel is now slightly bent using roller tool while it is turning on the machine. The edges at the opening of the vessel are removed by polishing and chiselling process.

Figure1: The Mandrel shaping the vessel

Figure 2: The Artisan shaping the vessel

After the desired shape is achieved, polishing is done using chemicals (Bright dipping) and then anodized/hard anodized.

4. GENERAL QUALITY CHECKS

Following are a few visual quality parameters/checks:

- **Fabrication quality:** Check for fabrication quality and workmanship
- **Shape and dimensions:** Check for shapes and dimensions
- **Proper finishing of anodized coating:** Check for finishing of anodizing coating and damages

5. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations, domestic and international (some key regions/countries), are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organisations (https://www.fieo.org/)
- Directorate General of Foreign Trade (https://www.dgft.gov.in/CP/)
- Export Promotion Councils (https://commerce.gov.in/useful-links/export-promotion-councils/)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce (https://commerce.gov.in/international-trade/trade-agreements/).

5.1. Domestic

S. No.	Country	Regulation	Regulatory Body	Remarks
1	India	a. Food Safety and Standards (Licensing and Registration of Food Businesses) Regulations, 2011	a. FSSAI	a. Governs use of Food grade utensils b. All equipment and utensils which come in contact with food.

5.2. International

S. No.	Region/ Country	Regulation	Regulatory Body	Remarks
1	European Union	a. Commission Regulation (EC) No. 2023/2006 of 22 December 2006 on good manufacturing practice for materials and articles intended to come into contact with food b. Framework Regulation (EC) No. 1935/2004	a. European Commission	a. Specifies requirements with respect to chemicals and heavy metals b. Specifies good manufacturing practices for materials coming in contact with food c. Specifies general safety requirements for all food contact materials. in addition, specific measures have been introduced to plastics
2	United States of America	a. Consumer Product Safety Improvement Act (CPSIA), Act of 2008. b. Consumer Product Safety Act (CPSA) of 1972.	a. California State b. Consumer Products Safety Commission (CPSC) c. Food and Drug Administration	a. Specifies requirements with respect to Chemicals and heavy metals b. Specifies requirement for suitable Product Labelling

S. No.	Region/ Country	Regulation	Regulatory Body	Remarks
		c. Federal Hazardous Substances Act (FHSA), Act of 1960. d. California Proposition 65, Act of 1986. e. Code of Federal Regulations, Section 21.		c. Specifies Banned substances d. Any manufacturing unit having more than ten employees are required to label products if they contain carcinogenic products e. Regulations of hazardous chemicals in Food contact materials (FCM)
3	Middle East	a. GSO Technical Regulations I. GSO 2231: 2012: 2012 II. GSO 839/1997	a. GCC Standardization Organization (GSO)	a. General requirements for the materials intended to come into contact with food

6. VOLUNTARY STANDARDS

Domestic Standards			International Standards		
Stage	**Code**	**Aspects Covered**	**Region/ Country**	**Standard**	**Aspects Covered**
General (Refer Table 1)	**IS 10259:1982 (Reaffirmed Year: 2020)**	General conditions of delivery of material	Middle East	**GSO 2231: 2012**	General requirements for the materials intended to come into contact with food
	IS 21:1992 (Reaffirmed Year: 2022)	Wrought Aluminium and Aluminium Alloys for Manufacture of Utensils	Middle East	**GSO 839: 2021**	Food packages - Part 1: General requirements
	IS 3544:2009 (Reaffirmed Year: 2019)	Glossary of terms	-	-	-
	IS 5047: PART1: 1986 (Reaffirmed Year: 2022)	Special types of wrought and unwrought aluminium	-	-	-
Raw Materials (Refer Table 2.a & 2.b)	**IS 737: 2008 (Reaffirmed Year: 2018)**	Chemical properties mechanical properties	European Union	**BS EN 851: 2014**	Utensils :Circle and circle stock
	IS 2676:1981	Dimensions and specifications for tolerances	USA	**ASTM E 716-16**	Chemical composition : Spark atomic emission spectrometry
	-	-	USA	**ASTM B209M-14**	Standards aluminium and aluminium-alloy sheet and plate
	-	-	Australia	**AS 4861 - 2004**	Tests for impurities in alloying elements
	-	-	Australia	**AS 2848.1-1998(R 2018)**	Designations for aluminium and its alloys
	-	-	Australia	**AS 1824**	Specifications for aluminium ingots

Domestic Standards			International Standards		
Stage	**Code**	**Aspects Covered**	**Region/ Country**	**Standard**	**Aspects Covered**
	-	-	Australia	**AS 3719.2**	Testing for silicon content
	-	-	Middle East	**GSO ISO 209: 2008**	Chemical composition of aluminium and alloys
	-	-	Middle East	**GSO ISO 797: 2015**	Gravimetric tests for silicon content
Manufacturing/ Processing (Refer Table 3.a & 3.b)	**IS 1868: 1996 (Reaffirmed Year: 2021)**	Suitable grades of aluminium for anodizing	European Union	**BS EN 12778: 2002**	Cookware, cutlery and flatware
	IS 1660: 2009 (Reaffirmed Year: 2021)	Thickness of aluminium sheets	USA	**AA SPC-2021**	Properties aluminium alloys used in casting
	IS 5893: 1989 (Reaffirmed Year: 2019)	Chemical composition of cryolite	Australia	**AS 1231-2000**	Anodic oxidation coatings and their standards
	-	-	Middle East	**GSO 1008: 1998**	Standards cold form household utensils
Finishing (Refer Table 4.a & 4.b)	**IS 5523: 1983 (Reaffirmed Year: 2021)**	Tests to measure thickness of anodized coating	European union	**BS EN ISO 84421-1:1997 (en)**	Table hollowware utilized for food preparation
	IS 7739: Part 3: 1975 (Reaffirmed Year: 2018)	Code of practice for preparation of metallographic specimens	European union	**BS EN 14372 - 2004**	Tests for childcare articles and feeding utensils
	IS 6507: 1991 (Reaffirmed Year: 2020)	Sodium Chromate Technical	USA	**ANSI HM35.2M-2017**	Dimensional tolerances
	IS 9040: 1978 (Reaffirmed Year: 2020)	Methods of sampling of utensils	-	-	-

6.1. Table 1: (Domestic & International General Standards)

Stage	Code	Scope
General	**IS 10259:1982 (Reaffirmed Year: 2020)** - Aluminium and its alloys, general conditions for delivery and inspection	▪ General conditions for delivery for Aluminium alloy sheets ▪ Mainly applicable to wrought alloys in as devolved conditions ▪ Tender Information, purchase order, ordering and Quality assurance ▪ Types of tests and their applicable standards
	IS 21: 1991 - Wrought aluminium and aluminium alloys for manufacture of utensils	▪ Chemical composition, and Mechanical properties of Wrought Aluminium and Aluminium Alloys for utensils ▪ Specification of Aluminium grades ▪ Testing Certificates. ▪ Shearing tolerances
	IS 3544:2009 (Reaffirmed Year: 2019) - Glossary of terms related to electroplating	▪ Glossary of terms related to electroplating

Stage	Code	Scope
	IS 5047: PART1: 1986 (Reaffirmed Year: 2022) - Glossary of terms related to aluminium and its alloys (wrought and unwrought)	▪ Main and Special types of wrought and unwrought aluminium ▪ Technical definitions of terms not including necessarily all legal meanings
	GSO 2231: 2012 – 2000 - General requirements for the materials intended to come into contact with food	▪ General requirements for Food contact materials
	GSO 839: 2021 - Food Packages - Part 1: General Requirements	▪ General requirements for all packages of food materials, including metal, glass, plastic, paper, carton, multi-layered textile, and wood packages, in addition to other materials for packaging foodstuffs

6.2. Table 2.a: (Domestic Standards for Raw Materials)

Stage	Code	Scope
Raw Material	**IS 737: 2008 (Reaffirmed Year: 2018)** - Alloy and wrought aluminium sheets	▪ Chemical composition of wrought and aluminium alloy sheets ▪ Mechanical properties ▪ Comparison of ISO and IS designations ▪ Characteristics and typical uses
	IS 2676:1981 - Wrought aluminium, aluminium alloys sheet and strips dimensions	▪ Dimensions for wrought aluminium and aluminium alloys. ▪ Specification of tolerances ▪ Thickness of wrought aluminium and aluminium alloy sheets. ▪ Shearing tolerances

6.3. Table 2.b: (International Standards for Raw Materials)

Stage	Code	Scope
Raw Material	**BS EN 851: 2014** - Aluminium and aluminium alloys. Circle and circle stock for the production of culinary utensils-Specifications	▪ Wrought aluminium and aluminium alloys for culinary utensils
	ASTM E 716-16 - Standard practices for sampling and sample preparation of aluminium and aluminium alloys for determination of chemical composition by spark atomic emission spectrometry	▪ Procedures for casting a chill cast disc from molten aluminium during the production process ▪ Procedures for producing a chill cast disk sample from molten metal produced by melting pieces cut from products.
	ASTM B209M-14 - Standard specification for aluminium and aluminium- Alloy sheet and plate (metric)	▪ Specifications Metric for aluminium and aluminium- Alloy sheet and plate (metric)
	AS 4861 – 2004 - Determination of impurities and alloying elements	▪ Determination of impurities and alloying elements
	AS 2848.1-1998(R 2018) - Aluminium and aluminium alloys – Compositions and designations	▪ Compositions and designations for aluminium and its alloys
	AS 1824 - Aluminium and aluminium alloys - Ingots and castings	▪ Aluminium and its alloys, Ingots and castings
	GSO ISO 209: 2008 - Aluminium and aluminium alloys - chemical composition	▪ Designation and chemical composition

Stage	Code	Scope
	GSO ISO 797:2015 - Aluminium and aluminium alloys - Determination of silicon - Gravimetric method	▪ Gravimetric method for the determination of silicon in aluminium and aluminium alloys. ▪ Applicability and Exclusions

6.4. Table 3.a: (Domestic Standards for Processing)

Stage	Code	Scope
Processing	**IS 1868: 1996 (Reaffirmed Year: 2021)** - Anodic coatings on aluminium and its alloys	▪ Guidelines on suitable grades of aluminium for anodizing ▪ Types of anodized aluminium ▪ Measurement practices and its definition ▪ Thickness of anodic oxide coatings and tests to check for resistance to corrosion, abrasion and crazing by deformation.
	IS 1660: 2009 (Reaffirmed Year: 2021) - Specifications for wrought aluminium utensils	▪ Requirements pertaining to quality and thickness of materials ▪ Thickness of base of various thick bottomed cooking utensils ▪ Workmanship and finishes of products
	IS 5893: 1989 (Reaffirmed Year: 2019) - Specifications for cryolite used in aluminium industry	▪ Bulk Density ▪ Chemical composition of cryolite ▪ Definition of referee method in case of disputes

6.5. Table 3.b: (International Standards for Processing)

Stage	Code	Scope
Processing	**BS EN 12778: 2002** - Specifications for cryolite used in aluminium industry	▪ Pressure cookers for domestic use is classified in these ICS categories**:** ○ 97.040.60 Cookware, cutlery and flatware ○ 97.040.20 Cooking ranges, working tables, ovens and similar appliances
	AA SPC - 2021 - Aluminium standards for sand and permanent mould casting	▪ Establishes matrices for tolerances, precision, allowances applicable to un-machined cast aluminium. ▪ Properties of various aluminium alloys used in casting and metallurgical processes
	AS 1231-2000 - Aluminium and aluminium alloys - anodic oxidation coatings	▪ Anodic oxidation coatings and tests
	GSO 1008: 1998 - Cold formed household utensils	▪ Cold formed household utensils ▪ Exclusions

6.6. Table 4.a: (Domestic Standards for Finishing)

Stage	Code	Scope
Finishing	**IS 5523: 1983 (Reaffirmed Year: 2021)** - Methods of testing anodic coatings on aluminium and its alloys	▪ Determination of thickness of anodized coating on aluminium and its alloys and methods of testing. ▪ Methods of determination of resistance to abrasion ▪ Methods of testing for light fastness and allocation of light fastness number ▪ Methods for testing of sealing
	IS 7739: Part 3: 1975 (Reaffirmed Year: 2018) - Code of practice for preparation of metallographic specimens	▪ Specifications of electrolytes for electro polishing. ▪ Hazards and safety protocols for handling electrolytes. ▪ Methods of testing for light fastness and allocation of light fastness number

Stage	Code	Scope
	IS 6507: 1991 (Reaffirmed Year: 2020) - Specifications for hard anodic coatings on aluminium and aluminium alloys	▪ Explanation of terminology ▪ Information to be given during placing of order to anodize ▪ Workmanship and finishes of products ▪ Thickness of coatings
	IS 9040: 1978 (Reaffirmed Year: 2020) - Methods for sampling of utensils	▪ Specification unified methods of sampling ▪ Criteria of acceptance for utensils ▪ Terms and terminology used ▪ Process inspection

6.7. Table 4.b: (International Standards for Finishing)

Stage	Code	Scope
Finishing	**BS EN ISO 84421-1:1997 (en)** - Methods of testing anodic coatings on aluminium and its alloys	▪ Cutlery and table hollowware in contact with foodstuff
	BS EN 14372 – 2004 - Code of practice for preparation of metallographic specimens	▪ Cutlery and feeding utensils. Safety requirements and tests is classified in these ICS categories: ○ 97.190 Equipment for children
	ANSI HM 35.2 M- 2017 - American national standard dimensional tolerances for aluminium mill products (metric)	▪ This standard covers the tolerances in metric units for sheet and plate

Note: The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

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    c. AUS: https://www.standards.org.au/
    d. GULF: https://www.gso.org.sa/store/
13. NABL accredited laboratories: https://nabl-india.org/, https://parakh.ncog.gov.in

BRASS UTENSILS, SANT KABIR NAGAR

1. INTRODUCTION

Brass is non-magnetic, a good conductor of heat, durable and easy to work on. All these properties make it ideal for being made into and used as utensils in kitchens.

The brassware of Sant Kabir Nagar is an ancient craft. The artisans engaged in this craft make various types of artistic utensils and pieces such as bowls, plates, glasses, vessels, jug, vase, bells and more. Brass Handicrafts produced here are very popular world over and they represent the true Indian craftsmanship and the old grandeur for which Indian Brass ware is well known. These exclusive and elegantly crafted brassware are not only popular in national markets but are also in huge demand in international markets.

Brass is an alloy of copper (60%-70%) and zinc (40%-30%). While copper is reddish in colour, brass is yellowish because of presence of zinc. Brass is more malleable that bronze or zinc which allows it to be easily drawn into sheets.

India and the world, is fast realising that traditional Indian cooking methods were perhaps the best for an individual's health. The types of utensils that were used in the earlier times, which were essentially metal or clay based, also contributed a lot to maintaining good health.

There is a growing disenchantment with the Teflon coated, plastic and other 'modern' utensils and their harmful effects on health are now well documented based on many research studies. The result of this growing awareness is that people are now increasing going back to traditional methods of cooking. For this reason, brass utensils are also being re-adopted as kitchenware.

2. RAW MATERIALS

The raw materials for making brass utensils are as below:

- Brass (Alloys of copper, silver) ingots or sheets
- Brazing flux powder

3. MANUFACTURING PROCESS

The process of manufacturing can either be done by hammering the strips into different shapes or by using moulds. Both of these processes are mentioned below:

3.1. Hammering and jointing of plates

a. **Cutting of the brass sheet:** The sheet of required thickness is procured from the market. It is first straightened and then with the help of a scale, marker and compass, markings are done on the metal sheet for the product to be made. In the next stage the sheet is cut either by chisel (if thick) or by metal scissor (if thin). The shape can be square or circular depending on the final product to be made. The thickness of the metal depends on the final product.

Figure 1: Cutting of brass sheet

b. **Shaping of the product:** Once the sheet is cut, it is placed on a 'Bangad' (Metal ring of iron) and hammered by wooden mallets into shape. Traditionally, 'Thiya' (Solid metal stand) is also used to shape the product or to make the neck of the vessels. While shaping the metal, the sheet is rotated continuously for even treatment of the shape of the metal body. For vessels such as pots, welding of the metal pieces made by hammering the sheet is needed.

Figure 2: Shaping of brass sheet

c. **Joining of pieces:** The products are made in parts and finally joined either by heating or by gas welding. Sometimes, artisans also join two pieces with traditional interlocking of the metals. These joints were converted into a design element by further heating and beating them. This process is carried out by heating the metal pieces in 'Bhatti' (furnace) at a high temperature. This makes the metal more flexible and thus merged/blended into each other. Today, artisans use gas welding as it is less time consuming and easy. For this process, both the pieces are heated till they become red hot. On the surface, 'suhaga' (a powder used for welding) is applied and a thin brass rod is melted which joins the two pieces. The gaps are filled by melting the brass rod on the product, thus helping in joining the two pieces.

d. **Finishing:** Once the product is joined, the excess metal accumulated on the surface due to welding is removed by grinding machine or files. Next, these items go through acid wash to clean the surface. The product is then buffed using buffs of different grades/numbers.

3.2. SAND BOX CASTING METHOD

a. **Making the mould:** The first step is to make the mould or 'master-piece' from which multiple products would be replicated (cast). These are usually made of wax, as it is soft and easy to work with. Sometimes, wood is also used. This 'master-copy' is always created in two (or more) detachable halves to make sand casting convenient.

Figure 3: Making of Mould

b. **Melting the metal:** The brass metal ingots or alloys (Copper, Zinc etc.) are heated in a furnace to melt them.

Figure 4: Pouring of brass

c. **Casting:** Sand casting is the traditional method of making brass ware. Sand is used in the two halves of the mould box to cast the metal which is prepared by packing sand (locally called 'masala') around the 'master-copy' tightly. The chemical binder in the sand aids in holding the shape of the mould after which it is removed and molten metal poured in its cavity. After being allowed to cool, the cast metal is removed from the mould box. The sand and the 'gating' (path way made in the mould for directing the flow of molten metal into the cavity) are broken from the cast using a hammer. These are re-used in the next casting.

d. **Scraping:** The products are mounted on a lathe and scraped to remove any protruding materials and to smoothen the surface. If the products are made in two parts (pots etc.) then the two pieces are joined by welding and then the product is subjected to scraping and polishing.

e. **Engraving:** Engraving is the most refined and artistic of all the processes. The design that has to be engraved is first sketched on paper and then scaled-up according to the size of the product. Measurement is an important aspect to make the pattern look harmonious. These are mainly inspired from different forms of nature like trees, flowers, birds and animals. The geometric ones have their influence from the Mughal architecture. First, an outline of the whole design is made with a fine engraving tool hammered with a wooden block and then, broader tools are used to engrave the background and give depth to the pattern. These are often filled with colourful lac or enamel.

Figure 5: Engraving

f. **Polishing:** Polishing mainly includes cleaning the brassware with a soft scrub and then buffing it on the machine for golden sheen.

g. **Qalai or Kalai:** For utensils to be used for cooking, a layer of tin is deposited to coat the surface. This is known as 'Qalai' locally. This is done as unprotected copper reacts with the moisture present in the air and forms copper carbonate (which can be seen on untreated copper vessels as light green rust on surface) which is poisonous and can be harmful if mixed with food. This layer of tin thus protects the copper vessel below. This however, wears off with time and as a result the copper vessels have to be repeatedly treated with the process of depositing a layer of tin. This is one of the reasons that has led to the decline in the use of copper vessels with the advent of aluminium and stainless-steel vessels.

4. GENERAL QUALITY CHECKS

Following are a few visual quality parameters/checks:

- **Proper finishing:** Check for uniformity in product or any other visual defects such as cracks, chipping, deformities etc.

- **Welding joints:** Check for smoothness & finishing of welding joints.

5. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations, domestic and international (some key regions/countries), are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organisations (https://www.fieo.org/)
- Directorate General of Foreign Trade (https://www.dgft.gov.in/CP/)
- Export Promotion Councils (https://commerce.gov.in/useful-links/export-promotion-councils/)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce (https://commerce.gov.in/international-trade/trade-agreements/).

5.1. Domestic

S. No.	Country	Regulation	Regulatory Body	Remarks
1	India	a. Food Safety and Standards (Licensing and Registration of Food Businesses) Regulations, 2011	a. FSSAI	a. Governs use of Food grade utensils b. All equipment and utensils which come in contact with food.

5.2. International

S. No.	Region/ Country	Regulation	Regulatory Body	Remarks
1	European Union	a. Commission Regulation (EC) No. 2023/2006 of 22 December 2006 on good manufacturing practice for materials and articles intended to come into contact with food. b. Framework Regulation (EC) No. 1935/2004.	a. European Commission	a. Specifies requirements with respect to Chemicals and heavy metals b. Specifies Good manufacturing practices for materials coming in contact with food c. Specifies General safety requirements for all food contact materials. In addition, specific measures have been introduced to plastics
2	United States of America	a. Consumer Product Safety Improvement Act (CPSIA), Act of 2008.	a. California State b. Consumer Products Safety Commission	a. Specifies requirements with respect to Chemicals and heavy metals b. Specifies requirement for suitable Product Labelling

S. No.	Region/ Country	Regulation	Regulatory Body	Remarks
		b. Consumer Product Safety Act (CPSA), Act of 1972. c. Federal Hazardous Substances Act (FHSA), Act of 1960. d. California Proposition 65, Act of 1986. e. Code of Federal Regulations, Section 21	c. Food and Drug Administration (FDA)	c. Specifies Banned substances d. Any manufacturing unit having more than ten employees are required to label products if they contain carcinogenic products e. Regulations of hazardous substances in food contact materials (FCM)
3	Middle East	a. GSO Technical Regulations I. GSO 2231: 2012: 2012 II. GSO 839/1997	a. GCC Standardization Organization (GSO)	a. General requirements for the materials intended to come into contact with food

6. VOLUNTARY STANDARDS

Domestic Standards			International Standards		
Stage	**Code**	**Aspects Covered**	**Region/ Country**	**Standard**	**Aspects Covered**
General (Refer Table 1)	**IS 3288: Part 1: 1986 (Reaffirmed Year: 2021)**	Glossary of terms relating to copper and copper alloys: Part 1 Materials	European Union	**BS EN 1655: 1997**	Copper and copper alloys. Declarations of conformity
	IS 1387: 1993 (Reaffirmed Year: 2018)	General requirements for the supply of metallurgical materials	-	-	-
Raw Materials (Refer Table 2.a & 2.b)	**IS 422: 1981 (Reaffirmed Year: 2021)**	Brass sheet and strip for the manufacture of utensils	European Union	**BS EN 1981: 2003**	Copper and copper alloys. Master alloys
	-	-	USA	**ASTM-B36/B36 M-2013**	Standard specification for brass plate, sheet, strip, and rolled bar
	-	-	USA	**ASTM B 455-01**	Standard Specification for Copper-Zinc-Lead Alloy (Leaded-Brass) Extruded Shapes
	-	-	Australia	**AS 1566 - 1997**	Copper and Copper alloys -Rolled flat products
Manufacturing/ Processing (Refer Table 3.a & 3.b)	**IS 4822: 1981 (Reaffirmed Year: 2021)**	Specifications for brass cooking utensils	European Union	**BS 3788**	Specification for tin coated finish on culinary utensils
	IS 3655: 1985 (Reaffirmed Year: 2021)	Recommended practices for electroplating	European Union	**BS EN 12167: 2011**	Copper and copper alloys. Profiles and bars for general purposes
	IS 16286: 2014 (Reaffirmed Year: 2020)	Spoon specifications with brass as parent material	European Union	**BS EN 1982: 2017**	Copper and copper alloys. Ingots and castings

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/ Country	Standard	Aspects Covered
	-	-	Middle East	**GSO ISO 10308: 2013**	Metallic coating
Finishing (Refer Table 4.a & 4.b)	**IS 3685: 1966 (Reaffirmed Year: 2018)**	Methods of chemical analysis of brasses	European Union	**BS EN ISO 6509-1: 2014**	Corrosion of metals and alloys. Determination of dezincification resistance of copper alloys with zinc - Test method
	IS 1501: Part 1:2020	Metallic materials — Vickers hardness Test Part 1 Test Method (Fifth Revision)	USA	**ASTM B605-95a**	Standard specification for electrodeposited coatings of tin-nickel alloy
	-	-	Australia	**AS 1515.4-1978 Rec - 2016**	Methods for analysis of copper alloys
	-	-	Australia	**AS 2345-2006**	Dezincification resistance of copper alloys
	-	-	Middle East	**GSO ISO 6509: 2013**	Corrosion of metals and alloys

6.1. Table 1: (Domestic & International General Standards)

Stage	Code	Scope
General	**IS 3288: PART 1: 1986 (Reaffirmed Year: 2021)** - Glossary of terms relating to copper and copper alloys	▪ Glossary of terms and technical definitions
	IS 1387: 1993 (Reaffirmed Year: 2018) - General requirements for the supply of metallurgical materials	▪ Raw materials supply for metallurgical industries ▪ Terminology and Inspection, sampling
	BS EN 1655: 1997 - Copper and copper alloys-Declarations of conformity	▪ Declaration of conformity, copper and copper alloys

6.2. Table 2.a: (Domestic Standards for Raw Materials)

Stage	Code	Scope
Raw Material	**IS 422: 1981 (Reaffirmed Year: 2021)** - Specification of brass sheet and strip for the manufacture of utensils	▪ Chemical composition ▪ Adjunct standards ▪ Physical properties

6.3. Table 2.b: (International Standards for Raw Materials)

Stage	Code	Scope
Raw Materi al	**BS EN 1981: 2003** - Copper and copper alloys	▪ Master Alloys**:** Copper and copper alloys

	ASTM-B36/B36 M-2013 - Standard specification for brass plate, sheet, strip, and rolled bar	▪ Requirements brass plates, sheets, strips, and rolled bars ▪ The standard tempers of as hot rolled, rolled, annealed, and annealed-to-temper materials are detailed in this specification ▪ Units of measurements

6.4. Table 3.a: (Domestic Standards for Processing)

Stage	Code	Scope
Processing	**IS 4822: 1981 (Reaffirmed Year: 2021)** - Specification for brass cooking instruments	▪ Requirements of brass cooking instruments ▪ Materials and dimensions
	IS 3655: 1985 (Reaffirmed Year: 2021) - Recommended standard practice for electroplating	▪ Recommended methods of practices for electroplating ▪ Process for parts in bulk ▪ Plating baths ▪ Adjunct standards
	IS 16286: 2014 (Reaffirmed Year: 2020)6 - Spoons specifications	▪ Requirements for the following types of spoons made from stainless steel and nickel silver plated with brass as a parent material

6.5. Table 3.b: (International Standards for Processing)

Stage	Code	Scope
Processing	**BS 3788** - Specification for tin coated finish on culinary utensils	▪ Specification for tin coated finish on culinary utensils
	ASTM B 455-01 - Standard specification for copper-zinc-lead alloy (leaded-brass) extruded shape	▪ Requirements for extruded leaded-brass angles, channels, and other architectural shapes of solid cross section produced in copper alloy ▪ Exclusions ▪ Units of measurements
	AS 1566 – 1997 - Copper and copper alloys - Rolled flat products	▪ Copper and copper alloys - Rolled flat products
	BS EN 12167: 2011 - Copper and copper alloys. Profiles and bars for general purposes	▪ Specifies the composition, property requirements and dimensional tolerances for copper alloy profiles including L-, T-, U-shaped cross-sections, and bars, finally produced by drawing or extruding
	BS EN 1982: 2017 - Copper and copper alloys. Ingots and castings	▪ Composition, mechanical properties and other relevant characteristics of copper and copper alloys ▪ Sampling methods ▪ Test Methods
	GSO ISO 10308: 2013 - Metallic coatings - Review of porosity tests	▪ Methods for revealing pores (see ISO 2080) and discontinuities in coatings of aluminium, anodized aluminium, brass, cadmium, chromium, cobalt, copper, gold, indium, lead, nickel, nickel-boron, nickel-cobalt, nickel-iron, nickel-phosphorus, palladium, platinum, vitreous or porcelain enamel, rhodium, silver, tin, tin-lead, tin-nickel, tin-zinc, zinc and chromate or phosphate conversion coatings (including associated organic films) on aluminium, beryllium-copper, brass, copper, iron, nifeco alloys, magnesium, nickel, nickel-boron, nickel-phosphorus, phosphor-bronze, silver, steel, tin-nickel and zinc alloy basis metal

6.6. Table 4.a: (Domestic Standards for Finishing)

Stage	Code	Scope
Finishing	**IS 3685: 1966 (Reaffirmed Year: 2018)** - Methods of chemical analysis of brasses	▪ Sampling ▪ Quality of reagents, Testing methodologies ▪ Test methodology
	IS 1501: Part 1:2020 - Vickers hardness test Part 1 Test method	▪ Hardness tests

6.7. Table 4.b: (International Standards for Finishing)

Stage	Code	Scope
Finishing	**BS EN ISO 6509-1: 2014** - Corrosion of metals and alloys. Determination of dezincification resistance of copper alloys with zinc Test method	▪ Corrosion of metals and alloys. Determination of dezincification resistance of copper alloys with zinc Test method
	ASTM B605-95a - Standard specification for electrodeposited coatings of tin-nickel alloy	▪ Standard specification for electrodeposited coatings of tin-nickel alloy
	AS 1515.4-1978 Rec – 2016 - Methods for the analysis of copper alloys	▪ Methods for the analysis of copper alloys. Electrolytic determination of copper in wrought and cast copper alloys ▪ Part 4: Electrolytic determination of copper in wrought and cast copper alloys
	AS 2345- 2006 - Dezincification resistance of copper alloys	▪ Dezincification resistance of copper alloys
	GSO ISO 6509: 2013 - Corrosion of metals and alloys	▪ Determination of dezincification resistance of brass. Applicable to brass exposed to fresh or saline water.

Note: The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

APPENDIX I: TESTING AND CERTIFICATION

Uttar Pradesh is home to a large number of indigenous products and manufacturing units associated with these unique products. The crafts and artisans have both withstood the vagaries of time and have flourished under the patronage provided in the form of various schemes to promote these products and bring about innovation in the industry by up skilling the artisans as well as providing them assistance in tapping new markets and becoming competitive in the modern market by utilizing machinery developed locally to augment their efforts.

With the advent of machinery and technology there has been a gradual move towards standardization to improve both the product life cycle as well as the quality of materials used. Two important aspects in standardization of products and processes are:

- Testing (Product/Material testing)
- Certification (Product Certification or Systems Certification)

1. PRODUCT TESTING

What is product testing?

Testing is the determination of one or more of a product's characteristics and is usually performed by a laboratory. In other words, product testing is a process to measure its performance, safety, quality, and compliance with established standards.

Quality testing is involved at each step of the production process and often begins with the testing of raw materials, moving through the testing of in-process products to the finished product.

Testing is an essential part of developing a quality product. Any product that is produced is expected to meet certain requirements and specifications to ensure standardization and consistency.

Why is testing of products required?

Testing is an essential part of developing a quality product. Product testing is required to assure quality, performance and safety of the product. Product testing is also required to comply with the regulatory requirements and standards, wherever prescribed.

Testing at the various stages of production helps to ensure that the products meet the defined specifications and gain acceptance globally.

Why should you get the product testing done from an approved/ accredited laboratory?

It is always recommended to get the tests done at a Laboratory that is approved /accredited by a competent authority. Accreditation indicates a formal recognition by an independent body, generally known as an accreditation body, which a certification body operates according to international standards.

Accreditation is an international network which is managed by the International Accreditation Forum (IAF), in the fields of management systems, products, services, personnel and other similar programmes of conformity assessment, and the International Laboratory Accreditation Cooperation (ILAC), in the field of laboratory and inspection accreditation. ILAC and IAF are international associations of accreditation bodies and they facilitate world trade through mutual recognition and also enhances the acceptance of products and services across national borders. Details on these international bodies and their MRA framework are available on www.iaf.nu and www.ilac.org, respectively.

In India, the National Accreditation Board for Testing and Calibration Laboratories (NABL), a constituent Board of Quality Council of India (QCI) has been established as the National Accreditation Body for accreditation of such laboratory. NABL is Mutual Recognition Arrangements (MRA) signatory to ILAC as well as APAC for the accreditation of Testing and Calibration Laboratories (ISO/IEC 17025), Medical Testing Laboratories (ISO 15189), Proficiency Testing Providers (PTP) (ISO/IEC 17043) and Reference materials producers (RMP).

Such MRA reduces technical barrier to trade and facilitates acceptance of test/ calibration results between countries which MRA partners represent.

Further, for certain products like food, pharmaceuticals etc., there are regulatory requirements (domestic as well as region or country specific (if exporting)) for testing of the raw materials and finished goods which are mandatory to be followed. For such products, various regulations and the standards making bodies like Bureau of Indian Standards (BIS) also notify specific Laboratories, from time to time. For e.g. BIS has recognised several laboratories for the tests prescribed under the Indian Standards formulated by them. Similarly, other regions or countries may have specific regulations that require prescribed tests, for e.g. REACH regulation of the European Union, specifies requirements to for the protection of human health and the environment from the risks that can be posed by chemicals.

Where can you get the list of accredited/approved laboratories?

List of the testing laboratories for the relevant tests may be obtained from the following URLs:

- **National Accreditation Board for Testing and Calibration Laboratories (NABL)**: https://nabl-Indiaorg/
- **Bureau of Indian Standards (BIS)**: https://bis.gov.in/

2. CERTIFICATION

Certification is a provision by an independent body of written assurance (a certificate) that the product, service or system in question meets specific requirements.

Broadly, certifications may be categorised as:

a. Management Systems Certification

These certifications help organizations improve their performance by specifying repeatable steps that organizations consciously implement to achieve their goals and objectives, and to create an organizational culture that reflexively engages in a continuous cycle of self-evaluation, correction and improvement of operations and processes through heightened employee awareness and management leadership and commitment.

The benefits of an effective management system to an organization include:

- More efficient use of resources and improved financial performance
- Improved risk management and protection of people and the environment
- Increased capability to deliver consistent and improved services and products, thereby increasing value to customers and all other stakeholders

Examples of Systems Certification include ISO 9001 QMS, ISO 22000 FSMS, etc.

b. Product Certification

Product certification is the process of certifying that a specific product fulfils requirements prescribed in contracts, regulations or specifications and has passed all necessary quality assurance and performance tests. Product certification is often required in industries and marketplaces where product failure could bring serious negative consequences to the health and safety of people and environment.

The benefits of an effective management system to an organization include:

- Product certification gives product users the confidence they need to purchase the desired products.
- If a product is certified according to applicable national or international laws, customers know that that product will function correctly and won't cause any harm. Examples of product certification include CE marking, ISI mark, Hallmark, etc.

Are all certifications mandatory?

Mostly, the certifications are voluntary in nature. However, for a number of products, compliance to specific Standards may be made compulsory, for instance, compliance to Indian Standards for a number of products, has been made compulsory by the Central Government under various considerations viz. public interest, protection of human, animal or plant health, safety of environment, prevention of unfair trade practices and national security. For such products, the Central Government directs mandatory use of Standard Mark under a Licence or Certificate of Conformity (CoC) from BIS through issuance of Quality Control Orders (QCOs). Similarly, other regions or countries may have specific regulations that require prescribed certifications, for e.g. USFDA prescribes specific requirements for Food, Drugs and Cosmetic products.

Why should you get the Certification from an accredited Certification Body?

It is always recommended to obtain the certification from accredited Certification Bodies or the ones that are prescribed by the relevant regulatory authorities. Accreditation indicates a formal recognition by an independent body, generally known as an accreditation body, which a certification body operates according to international standards.

Accreditation is an international network which is managed by the International Accreditation Forum (IAF), in the fields of management systems, products, services, personnel and other similar programmes of conformity assessment, and the International Laboratory Accreditation Cooperation (ILAC), in the field of laboratory and inspection accreditation. ILAC and IAF are international associations of accreditation bodies and they facilitate world trade through mutual recognition and also enhances the acceptance of products and services across national borders. Details on these international bodies and their MRA framework are available on www.iaf.nu and www.ilac.org, respectively.

In India, the National Accreditation Board for Certification Bodies (NABCB), a constituent board of Quality Council of India (QCI) has been established as the National Accreditation Body for accreditation of such Certification and Inspection Bodies. The National Accreditation Board for Certification Bodies provides accreditation to Certification and Inspection Bodies based on assessment of their competence as per the Board's criteria and in accordance with International Standards and Guidelines.

NABCB is internationally recognized and represents the interests of the Indian industry at international forums through membership and active participation with the objective of becoming a signatory to international Multilateral / Mutual Recognition Arrangements (MLA / MRA).

Where can you get the list of accredited/approved laboratories?

List of the NABCB accredited Certification Bodies (CBs) and Inspection Bodies (IBs) for relevant certifications may be obtained from the following URL:

National Accreditation Board for Certification Bodies (NABCB): http://nabcb.qci.org.in/

APPENDIX II: IMPORTANT INSTITUTIONS

UTENSILS				
S. No	**Institute**	**Areas of Expertise**	**Postal Address**	**Contact**
1	Process and Product Development Centre (PPDC)	▪ Material Testing ▪ Consultancy ▪ Product Development ▪ Training	Process and Product Development Centre (PPDC), Foundry Nagar, Agra, Uttar Pradesh - 282006	Email: tcppdcagra@dcmsme.gov.in Phone: +91–562- 2344673 Website: http://www.ppdcagra.dcmsme.gov.in/

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